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A MINERAGRAPHIC EXAMINATION OF HIGH BISMUTH PANNED
CONCENTRATE FROM N300/300 DRILL-HOLE AT 43.5M,
CATTLE GRID.

1. INTRODUCTION

A specimen of panned concentrate was submitted by Mr. A. Turnbull. The bismuth content was alleged to be between 5 and 7%. A polished section containing all the concentrate was prepared and examined.

2. SUMMARY

Wittichenite occurs as fine inclusions or as network intergrowths in many chalcocite grains. However, by optical examination alone there would not be anywhere near 5 to 7% bismuth in the sample as was indicated on the sample envelope.

3. INVESTIGATION

About 99 volume percent of the sulphide present consists of chalcocite fragments (90 volume percent blue variety and 9 percent bluish white. The median size of the grains is 0.030mm but with a maximum 0.090mm. Very few free sulphide fragments of less than 0.005mm size are seen. Minimal bornite and less sphalerite form complex intergrowths with the chalcocite, as do traces of pyrite and galena. Many pyrite and galena fragments however are free from other sulphide composites.

About 10 volume percent of mainly bluish chalcocite grains contain an abundance of small ovoid, lobate inclusions or intergrowths of wittichenite ($3 \text{ Cu}_2\text{S} \cdot \text{Bi}_2\text{S}_3$). No wittichenite grains free from composites are seen. The individual wittichenite grains vary in size down to 0.001mm, and are creamy white and distinctly anisotropic. The grains are difficult to see under low power magnification (X60 diameters) and are better seen using higher power magnification using oil immersion.

P.J. Curtis
(IM)

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PACMINEX PTY LIMITED

PETROGRAPHIC EXAMINATION OF ⁷McLEAY BEDS,

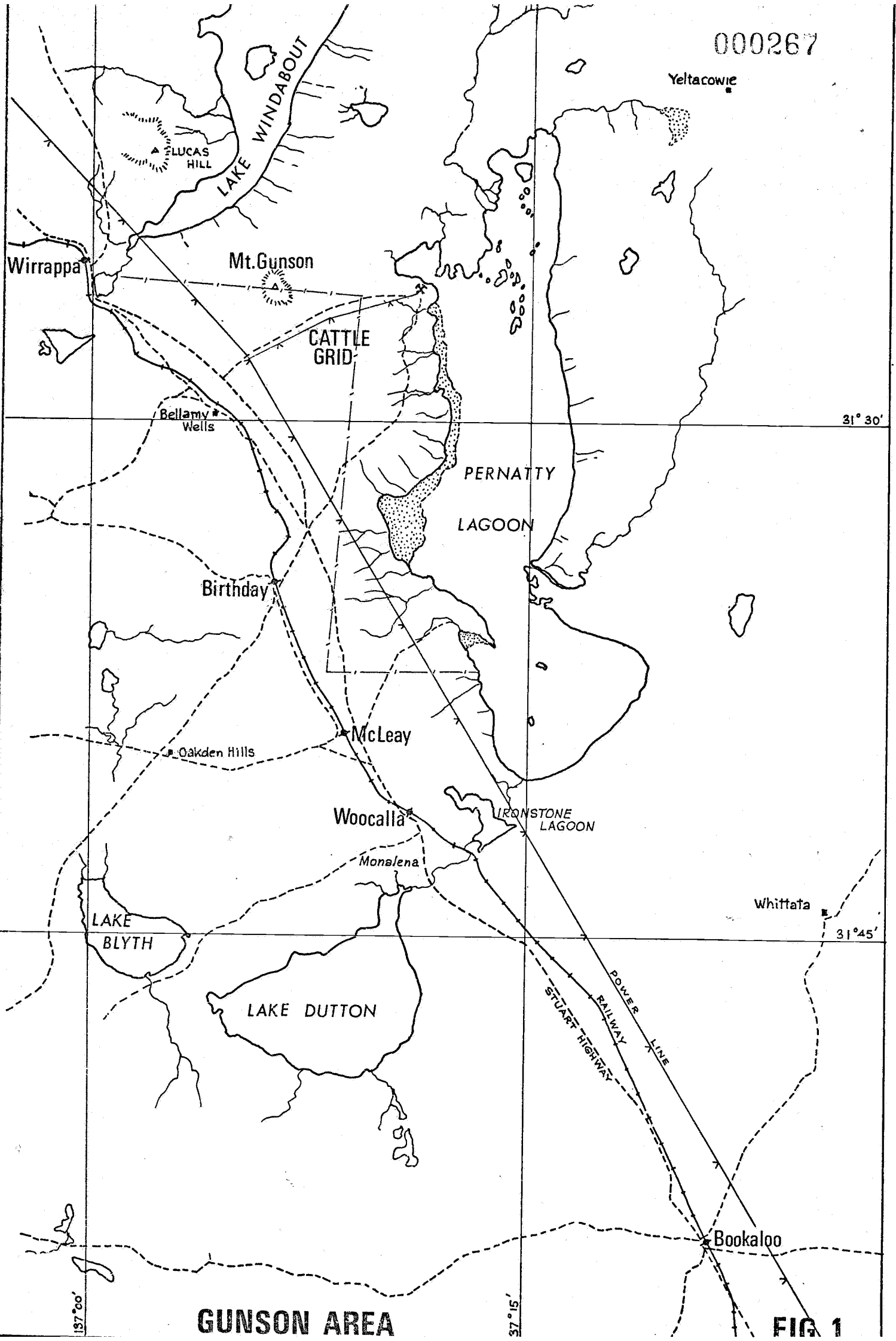
MT. GUNSON DISTRICT

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000267

Yeltacowie



GUNSON AREA

FIG 1

PACMINEX PTY. LIMITED

MEMORANDUM TO: Mr. P.J. Curtis

OUR REF: KJM/SS

FROM: K.J. Maiden

DATE: 9th January, 1974.

Typed: 13/3/74

Rocks for Petrological Examination

Would you please examine the enclosed rocks. They are from the "McLeay Beds" above the Pandurra Formation in the vicinity of McLeay railway siding.

- 71450 (A9501): Ferruginous deposit (with pebbles and fragments of quartzite) capping Pandurra Formation. This may be a surface-silicified version of the haematite-rich clay and sand above the Pandurra Formation in drill hole MG 14 D (sample 80126) A9967.
- 71451 (A9502): Pink quartzite containing angular fragments of red quartzite. This is similar to the pink quartzite at the top of the Pandurra Formation in MG 14D (sample 80123) A9966.
- 71452 (A9503): "Bedded" quartzite, finer grained than 71451.
- 71453 (A9504): "Green" quartzite - dark coloured laminated rock with cavities.
- 71454 (A9505): Grit - quartz pebbles and coarse sand.
- 71455 (A9506): Pebble conglomerate - similar to Elizabeth River conglomerate (A9862?)

K.J. MAIDEN

1. INTRODUCTION

The McLeay Beds constitute a stratigraphic transition zone between Pandurra Formation and overlying Whyalla Sandstone at McLeay railway sidings 21 km south-west of Mt. Gunson.

Six specimens were submitted for examination.

2. SUMMARY

A9501 (71450)

A relatively small number of large spherical quartz grains occur, but larger more angular grains of polymict origin are common. A two population distribution of quartz rock particles with more rounded silt-size particles are seen. Resemblance to sample A9967 (80126) is not good but there is a better resemblance to specimen P.39 from the Pandurra District (Ref. 1), though to be a ferruginous Whyalla Sandstone. However, the larger more angular grains and the low abundance of heavy minerals and ferruginous cherts place the rock outside standard Whyalla Sandstone classification.

A9502 (71451)

The rock has a distinctly upper (?) Pandurra Quartzite appearance and is somewhat similar to specimens selected from West Lagoon, Woocalla Quarry, etc.

A9503 (71452)

The rock consists of mainly subangular even-grained quartz and contains appreciable ferruginous cherts and heavy minerals. It does not completely fit in with either Whyalla Sandstone or Pandurra Quartzite types but has some properties of each of the stratigraphic units.

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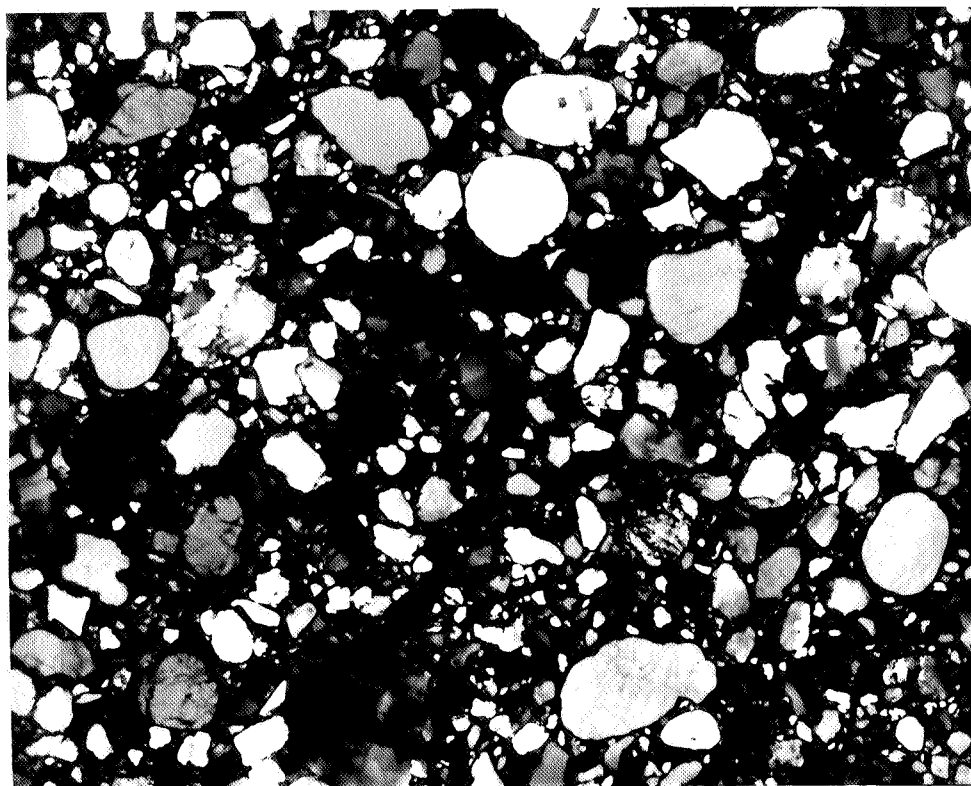


Figure 1: McLeay Beds (A9501/71450)

Large spherical quartz grains, subangular quartz and chert in silt-size quartz and limonitic matrix.

Crossed nicols

Magnification X14

A9504 (71453)

The heavy oxidation and abundant chert matrix set this rock apart from any sandstone seen so far from the area. The chert bears a resemblance to the chert (clear and ferruginous) commonly seen in Whyalla Sandstone at Mt. Gunson and it is suggested that possibly much of the Whyalla Sandstone chert originated from such a source.

A9505 (71454)

Not unlike very coarse Pandurra Quartzite from some Mt. Gunson occurrences. A silicified gravel. However, thin-sections of such a coarse Pandurra Quartzite are not available.

A9506 (71455)

Coarse polymict clasts and fine quartz matrix. The Elizabeth Creek specimen (Ref. 3) was regarded as Whyalla Sandstone with a ferruginous cement but this specimen (A9506) has several of the characteristics of Pandurra Quartzite.

3. PETROGRAPHIC EXAMINATION

A9501 (71450) (Figures 1, 2 and 3)

A ferruginous deposit (with pebbles and fragments of quartzite) capping Pandurra Formation. An iron stained silicified sandstone with many spherical quartz pebbles of variable size as seen in hand specimen.

In thin-section the overall proportion of spherical quartz, mineral and rock fragments is around 7 volume percent of grains present and includes some clear chert. The size range is 0.3 to 1.5 mm, median of approximately 0.6 mm. Most round grains are of mineral quartz. Less than 1 percent of the rock consists of spherical rock fragments and chert. Larger angular, flat, sub-angular and rounded mineral grains and quartz rock fragments (including quartz gneiss) up to 3 mm are occasionally seen and once again these

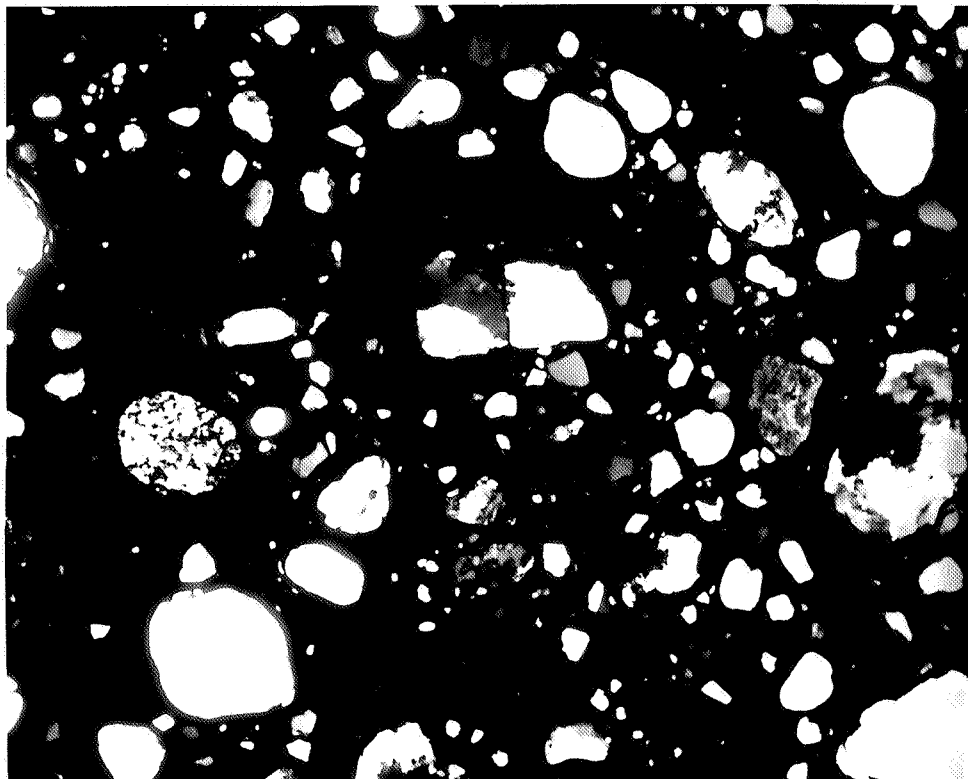


Figure 2: Ferruginous Whyalla Sandstone, Pandurra District; P.39

Rounded and subangular quartz and chert grains in a ferruginous, silty (obscured), matrix. Somewhat like A9501 (Figure 1) but matrix is more ferruginous.

Crossed nicols

Magnification X14

consist of a much greater proportion of quartz mineral grains, the whole constituting no more than 2 percent of the rock.

Much finer, round to angular (mainly rounded-subangular) grains of quartz mineral constitute most of the remaining grains. The range is 0.03 to 0.8 mm, median sizes (two population distribution) are 0.08 mm and 0.3 mm with a fairly sharp division between the two distributions.

Between the quartz grains is a heavy limonite iron oxide with less chalcedony cement. The limonite averages +0.05 mm thick and encroaches on adjoining quartz. Heavy mineral seen in the matrix is only epidote as traces. Chert grains of all kinds occur to about 2 volume percent. Volcanic ferruginous chert grains are present to less than one percent, most of these are iron oxide stained cherty-, quartz and one large round grain is of iron oxide stained, silicified granophyre.

The angularity of some of the larger grains, presence of occasional spherical quartz grains and also that of a silt size quartz matrix show some resemblance of this rock to A9967 (80126) from D.H. MG 14 (Ref. 1), but the larger grains are more angular and there are more rock fragments in the latter. There is a greater resemblance to P.39 from the Pandurra District (Ref. 2) thought to be a ferruginous variety of Whyalla Sandstone. Certainly the main fabric of A9501 has a vague Whyalla Sandstone appearance (after some Whyalla Sandstone occurrences from Mt. Gunson), with large round grains and the presence of silt quartz. However A9501 does have a greater proportion of large subangular quartz grains and the heavy mineral content and ferruginous volcanic chert content are low for a 'standard' Whyalla Sandstone.

A9502 (71451)

A pink quartzite which contains angular fragments of red quartzite. In hand specimen the pink colour and erratics of rock fragments give the rock a distinctly Pandurra Quartzite appearance.

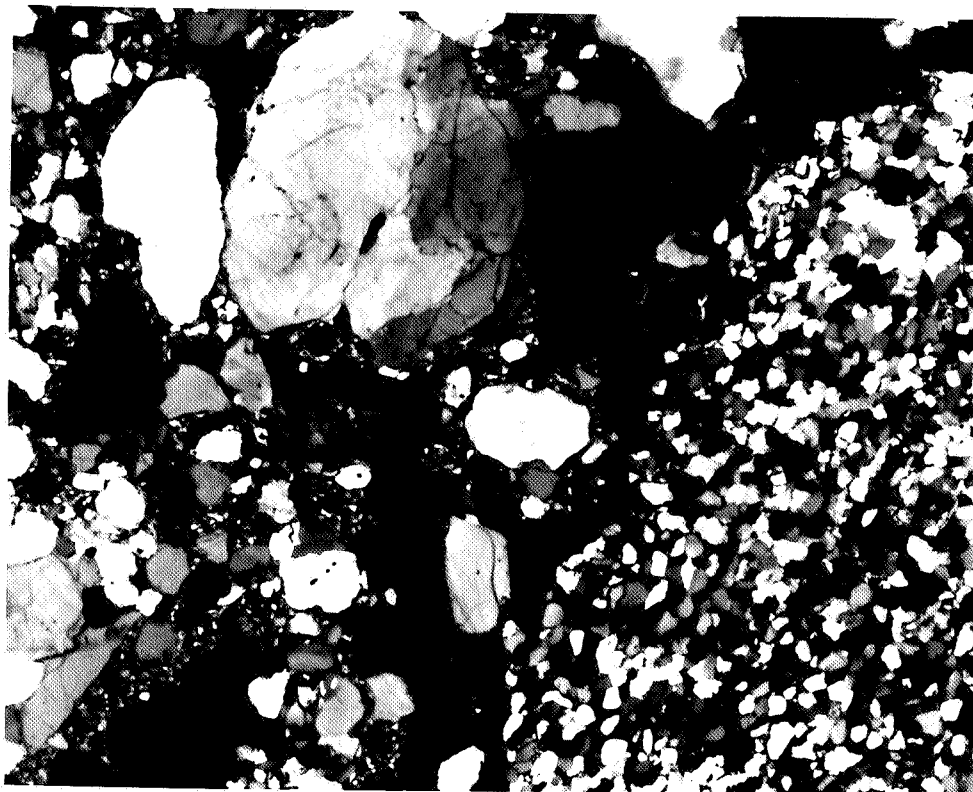


Figure 3: Quartzite fragments in clay matrix and silt.
A9967/80126, 36.50 m. DH MG 14D

Thought to be unsilicified version of A9501/71450. The matrix is similar in containing silt but larger rock fragments present are more irregular in size as seen in the photograph. One large rock fragment of fine quartzite occurs partly on right of photograph.

Crossed nicols

Magnification X14

In thin-section the rock is seen to be a silicified quartzite of mainly subangular or rounded subangular grains varying between 0.2 and 1.5 mm with a median grain size of 0.4 mm. Many grains were made subangular by silicification and originally more regular grains are present as rounded or oblong nuclei demarcated by hematite particle and fluid inclusion trains from authigenic overgrowths.

One large quartz rock fragment erratic, several millimetres in length and width, contains quartz grains of similar size and shape to the host rock but the cement between the grains is chalcedonic silica and hematite/limonite, the latter being more abundant than is seen in the host rock interstices. Groups of heavy minerals, topaz, sphene and epidote also occur.

Another fragment of red quartzite present shows abundant chert and hematite between larger quartz grains and also the presence of subangular silt-size quartz matrix.

One other fragment of red quartzite shows the presence of heavy minerals of magnetite with less zircon and sphene which are partly obscured by limonite/hematite matrix as seen in a red quartzite fragment in A9966, but the matrix in A9966 is illite/kaolinite where not silicified with quartz. The overall textures, grain size and mineral composition of A9966 (Pandurra Quartzite - Ref. 1) are in general very similar to A9502 as seen in thin-section and both specimens contain the quite large erratics.

The size and shapes of quartz grains and distribution of erratics are very similar to Pandurra Quartzite specimens from A4009 (West Lagoon), A4226, A4230 (Woocalla Quarry), A4250 (Pernatty Lagoon) and A4254 (Main Open Cut), (Ref. 4). However, A4250 has a cherty cement and A4254 retains abundant kaolinite in the interstices. Also the very large (several mm's) clasts of quartz rock are not apparent in thin-section in any of these specimens.

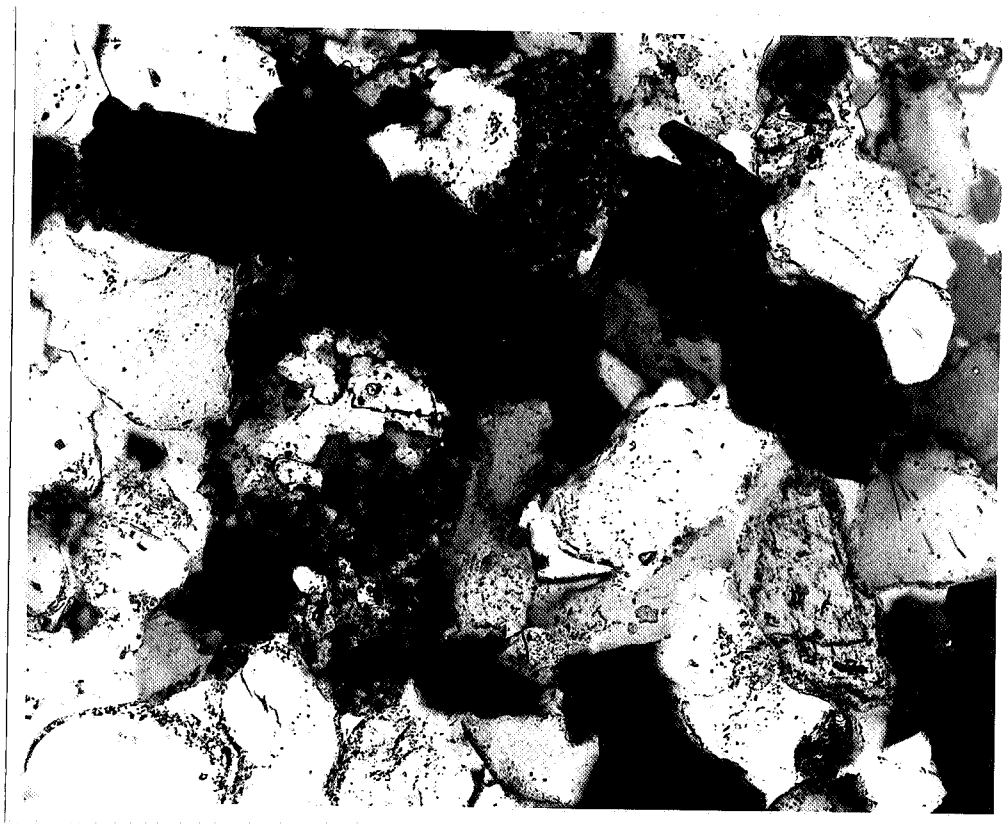


Figure 4: Rock with both Pandurra Formation and Whyalla Sandstone characteristics. A9503/71452.

Subangularity and even nature of quartz grains noted. Abundance of chert as small limonitic mosaics and speckled black and white grains (left and upper side of photograph). Sphene is seen (near upper, right corner).

Crossed nicols

Magnification X100

In conclusion, this rock is very similar to some other Pandurra Quartzite rocks examined, probably texturally resembling Pandurra Formation from close to the top of the formation most of all.

A9503 (71452) (Figure 4)

Bedded quartzite, finer grained than A9502. Fine sugary pink quartzite specked with opaques in places. Surface of the rock is pitted.

The rock is seen to consist of even-grained mainly subangular quartz grains. Many quartz grains (range 0.03 to 0.4 mm, median 0.2 mm) show overgrowths, the rock being completely silicified; occasional large erratics of quartz of approximately 1 mm size occur as quite rounded grains. Up to half the quartz grains were originally quite round or oval as indicated by internal narrow trains of hematite particles and fluid inclusions. Silicification of rounded silt-size quartz grains is seen locally. Minimal interstitial chert occurs in pockets and channels. Rounded ferruginous cherts and sericitic ferruginous cherts (range in size, 0.1 to 0.8 mm), as are normally common in Whyalla Sandstone are present to 3 volume percent.

Scattered, rounded zircons ($\sim$ 0.10 mm) occur to less than 1 volume percent of rock and less rounded green tourmaline and sphene of similar or slightly larger grain size are seen. Rounded vugs ($\sim$ 0.3 mm) are common to 3 volume percent of the rock. The abundance of originally rounded quartz, ferruginous cherts and appreciable although not abundant heavy minerals are not typical characters of Pandurra Quartzite but are characteristic of Whyalla Sandstone. The evenness of grains with sporadic quartz erratics is typical of some Pandurra Quartzite fabrics.

The quartzite of this rock is typical of some clasts seen in Pandurra Quartzite (as in A9967 - (80126) from DH MG 14D); but in A9503, polymict clasts are not seen and this includes ferruginous quartzite clasts; heavy minerals are present and there is no interstitial clay mineral in A9503.

From the above it is seen that this rock does not quite fit with the accepted textural and mineral composition assumption for typical Pandurra Quartzite.

A9504 (71453)

'Green' quartzite - dark coloured laminated rock with cavities.

Mainly subangular, medium size quartz in a dark limonite and chert matrix. Large rounded and subangular quartz erratics (~ 1 mm) occur in a fairly even-grained mass of quartz grains (0.1 to 0.6 mm, median 0.4 mm). Silicification by chert has taken place and this gives the rock the green colour. Also broad bands of limonite particles occur just within the periphery of many quartz grains and particularly where they adjoin the chert matrix. The chert matrix constitutes about 10 volume percent of the rock and forms broad parallel laminae up to 1 mm or more in width containing islands of quartz grains. Some of this chert has a dark limonite stain. Occasional, (< 1 volume percent of rock) clear detrital cherts are seen. These tend to be rounded and are of similar grain size to the neighbouring quartz grains.

Some of the clear and ferruginous (non-volcanic) chert of the Whyalla Sandstone may have originated from this horizon. There are definite similarities between interstitial chert seen here to many of the chert particles observed commonly in the Whyalla Sandstone. The rock has been weathered to the extent of Pandurra Quartzite in the Main Open Cut Gunson which also has abundant cherty matrix.

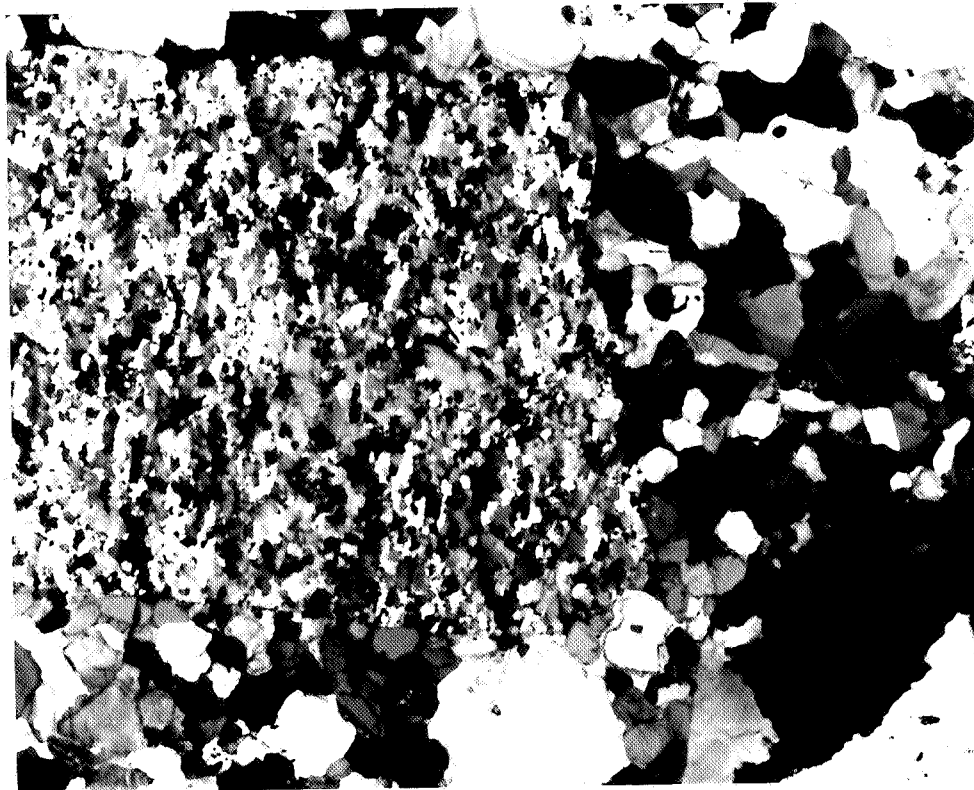


Figure 5: Rock supposedly similar to Elizabeth River conglomerate. A9506/71455.

Large fragment of fine grain quartzite to left of photograph and typical Pandurra Formation mosaic to right.

Crossed nicols

Magnification X14

000280

There is a little 'interstitial' lozenge shaped sphene (~ 0.6 mm) present but it is rarely clear and tends to be obscured by deep limonite staining. Other heavy minerals may be present, apart from sphene and limonitised hematite crystal groups but they are obscured by the deep stain. Zircons present (< 1 volume percent) are rounded (size ~ 0.80 mm).

A9505 (71454)

Grit-quartz pebbles and coarse sand. An irregular grit containing conglomerate lenticles.

Not unlike Pandurra Quartzite on a coarser scale. Poorly sorted quartz grains (range 0.2 to 8.0 mm). Many of the larger grains are quite angular and are quartz rock fragments. The smaller grains (mainly quartz mineral), show rounded grain nuclei with more angular overgrowths, the original grains being outlined by fine particle hematite inclusions trains. Occasional large quartz gneiss grains are seen. Clear chert grains occur to 2 volume percent and are commonly relatively small (~ 0.6 mm) and rounded. Ferruginous cherts are of similar size and shape to clear cherts and occur to about 1 volume percent of the rock. Occasional cavities occur between larger quartz grains which may have contained clay mineral which was lost in section preparation, otherwise commonly the rock is a silicified quartz mosaic. Rare, rounded zircon grains (0.03 mm) are seen embedded in quartz overgrowths. No quartz silt-size material occurs and no ferruginous cherts with volcanic textures are seen. Heavy minerals are rare.

A silicified gravel.

A9506 (71455) (Figures 5 and 6)

Pebble conglomerate which is allegedly similar to Elizabeth River conglomerate. Hand specimens show a two population distribution conglomerate of large (several millimetres diameter) rounded quartz fragments and a much finer quartz matrix.

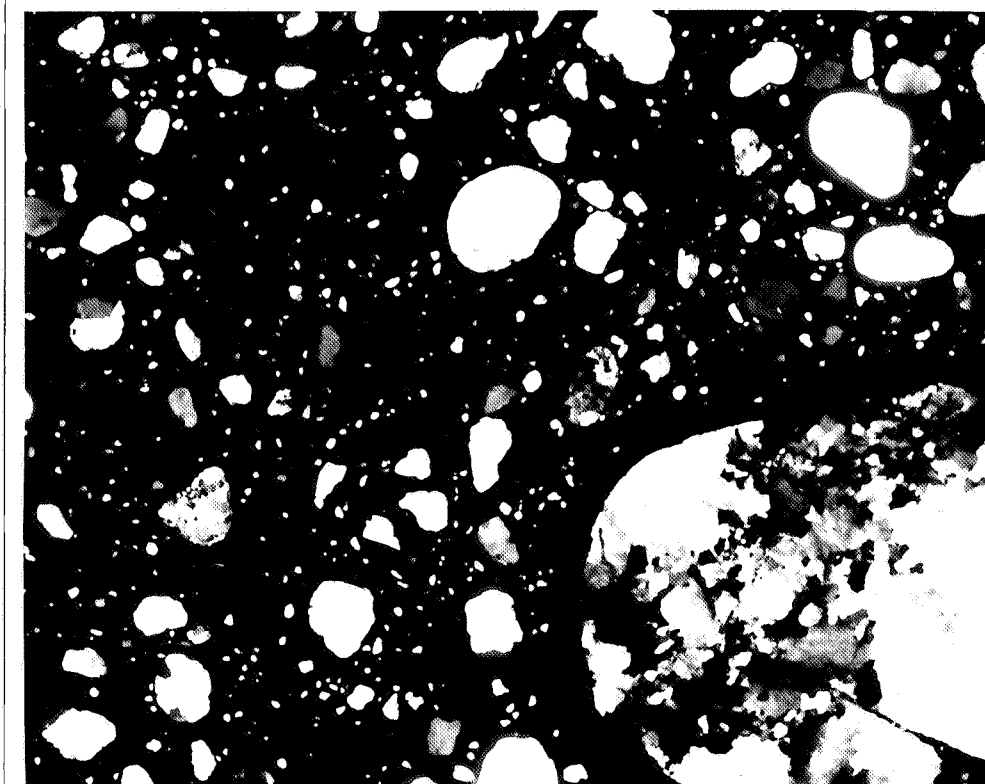


Figure 6: Whyalla Sandstone with ferruginous cement described as Elizabeth River conglomerate. A9862/70743.

One large spherical clast (right bottom of photograph). Silt size quartz and limonite constitute the matrix. Quite unlike A9506/71455 specimen, (Figure 5).

Crossed nicols

Magnification X14

In thin-section some large quartz rock fragments are seen to be not so rounded and are cherty looking. However, most large grains are rounded but are not spherical; two fragments are distinctly triangular in shape. The source of the large grains is variable, some being gneissose metamorphic. Others are coarse quartzite fragments with intricately sutured quartz grains; yet others consist of very fine even-grain quartzite as seen in some clasts of Pandurra Quartzite and in the fabric of A9503 of this series. The rock matrix consists of typical even-grained Pandurra Quartzite where this occurs, of quartz grains 0.2 to 1.0 mm but mainly close to median size 0.3 mm. Many of these grains were quite rounded but quartz overgrowths have made the grains somewhat subangular and have produced a silicified mosaic but with no intricate suturing between quartz grains as is seen in some of the large clasts. Small rounded ferruginous cherts (~0.3 mm) which occur to about 1 volume percent of the rock are rounded but no volcanic textures are seen. One large subangular ferruginous chert (10 mm in width) occurs. Apart from hematite (after magnetite?) grains in one of the large cherty clasts and occasional specks of opaque mineral in small ferruginous cherts there are no heavy minerals to be seen in this rock.

In A9862 (70743), Whyalla Sandstone described as a ferruginous deposit from Elizabeth Creek, the large clasts are several millimetres in diameter but are quite spherical in shape and are smaller than the clasts seen in A9506. However, there is a silt-size quartz where seen through the dense limonite cement but again these quartz grains are smaller than those in the matrix of A9506.

There are few similarities between the Elizabeth Creek, Whyalla Sandstone A9862 and the rock A9506 presently discussed. A common origin of some grains (cherts, gneiss fragments, etc.) is suspected.

5. REFERENCES

1. Memo from P.J. Curtis to K.J. Maiden dated 8th January, 1974, entitled 'Samples from drill-hole MG 14D containing copper mineralisation in fine dolomite.'
2. Pacminex Pty. Limited Report No. PMR 5/74 dated January, 1974, entitled 'Petrography of Pandurra Thin Sections, II'
3. Memo from P.J. Curtis to K.J. Maiden dated 15th October, 1973, and entitled 'Rocks for Petrological-Mineralogical Examination.'
4. Pacminex Pty. Limited Report No. PMR 195/72 dated October, 1972, and entitled 'Petrography of Mount Gunson sediments.'
5. Memo from K.J. Maiden to P.J. Curtis dated 9th January, 1974, entitled 'Rocks for Petrographic Examination.'